

AI23331 Fundamentals of Machine Learning

AIR QUALITY INDEX

A PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

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ABSTRACT

Air pollution has become a critical environmental and public health concern in urban and industrial regions. This project presents an intelligent air quality monitoring and prediction system that integrates machine learning with real-time data visualization. The system utilizes pollutant parameters such as PM_{2.5}, PM₁₀, NO₂, CO, SO₂, and O₃ to analyze air quality conditions and predict the Air Quality Index (AQI).

A Random Forest Regressor model is trained on historical air quality data to accurately estimate AQI values based on pollutant concentrations. The model performance is evaluated using standard metrics such as Mean Absolute Error (MAE) and R² score. In addition, a comparative analysis with baseline models such as Linear Regression is performed to ensure optimal model selection.

The developed system is deployed using a Streamlit-based web application, which allows users to input a city name or select a geographic location through an interactive map. The application fetches real-time pollution data using an external API and provides both live AQI values and machine learning-based predictions. It also includes graphical visualization of pollutant levels, AQI categorization, and health advisory recommendations for users.

This project demonstrates the practical application of machine learning in environmental monitoring and highlights the importance of data-driven approaches

for pollution management and public awareness. The system aims to assist individuals and authorities in making informed decisions to reduce exposure to harmful air pollutants.

1. INTRODUCTION

Air pollution is one of the most serious environmental challenges faced by modern society, particularly in rapidly urbanizing and industrializing regions. The increasing concentration of harmful pollutants such as particulate matter (PM_{2.5}, PM₁₀), nitrogen oxides (NO₂), carbon monoxide (CO), sulfur dioxide (SO₂), and ozone (O₃) has led to significant impacts on human health, climate change, and overall environmental quality. Prolonged exposure to polluted air can result in respiratory diseases, cardiovascular problems, and reduced life expectancy.

Monitoring and managing air quality has therefore become essential for ensuring public safety and sustainable development. Traditional air quality monitoring systems rely on fixed stations and often provide delayed or limited information. With the advancement of data science and machine learning techniques, it is now possible to analyze large volumes of environmental data and generate accurate predictions of air quality levels in real time.

This project aims to develop an intelligent air pollution monitoring and prediction system that combines machine learning with real-time data acquisition. A Random

Forest Regression model is used to predict the Air Quality Index (AQI) based on various pollutant concentrations. The system also integrates real-time data through external APIs and presents the information using an interactive web application built with Streamlit.

The proposed solution not only provides current air quality conditions but also predicts future AQI levels and offers health advisory recommendations. By enabling users to input a city or select a location on a map, the system enhances accessibility and usability. This project demonstrates how machine learning and modern web technologies can be effectively utilized to address real-world environmental problems and promote awareness about air pollution.

2. LITERATURE REVIEW

2.1 Machine Learning for Air Quality Prediction

Air quality prediction has become an important research area due to increasing environmental concerns and public health risks. Traditional statistical methods were initially used to estimate pollution levels, but they often failed to capture complex nonlinear relationships among pollutants. With the advancement of machine learning, models such as Random Forest, Support Vector Machines, and Gradient Boosting have been widely adopted for AQI prediction. Among these, Random Forest Regressor has shown strong performance due to its ability to handle nonlinear data, reduce overfitting, and provide reliable predictions based on multiple decision trees.

2.2 Time Series and Deep Learning Approaches

Air pollution data is inherently time-dependent, which led researchers to explore time series models such as ARIMA and deep learning techniques like Long Short-Term Memory (LSTM) networks. LSTM models are capable of capturing temporal dependencies and trends in pollution data, enabling more accurate forecasting of

AQI over time. More recent approaches include hybrid models combining CNN and LSTM architectures, which can extract both spatial and temporal features from environmental data.

2.3 Feature Engineering and Pollutant Analysis

Effective AQI prediction relies heavily on selecting relevant input features such as PM_{2.5}, PM₁₀, NO₂, CO, SO₂, and O₃ concentrations. Researchers have emphasized the importance of feature engineering techniques, including handling missing values, normalization, and correlation analysis. Studies show that particulate matter (PM_{2.5} and PM₁₀) plays a dominant role in determining AQI levels, making them critical inputs for predictive models.

2.4 Real-Time Air Quality Monitoring Systems

With the growth of IoT and smart city technologies, real-time air quality monitoring systems have gained popularity. These systems integrate sensor data, satellite observations, and external APIs to provide live pollution information. Platforms such as OpenWeatherMap and other environmental data services allow applications to fetch real-time pollutant data, enabling dynamic monitoring and immediate analysis of air quality conditions.

2.5 Visualization and User-Centric Applications

Data visualization plays a key role in making air quality information understandable to users. Graphical representations such as bar charts, line graphs, and geographic maps are commonly used to display pollutant concentrations and AQI levels. Interactive web applications built using frameworks like Streamlit have made it easier to present complex environmental data in a user-friendly manner, allowing users to explore air quality information efficiently.

2.6 Integration of Machine Learning with Web Applications

Recent research focuses on integrating machine learning models with web-based interfaces to create intelligent decision-support systems. Such systems combine predictive analytics with real-time data retrieval, enabling users to obtain both current and forecasted air quality information. This integration enhances usability and ensures that machine learning models are accessible to non-technical users through interactive dashboards.

2.7 Health Impact Assessment and Advisory Systems

Air quality systems are increasingly incorporating health advisory mechanisms based on AQI levels. These systems classify air quality into categories such as Good, Moderate, Poor, and Severe, and provide recommendations accordingly.

This approach helps individuals take preventive measures, such as reducing outdoor activities or using protective masks, thereby minimizing exposure to harmful pollutants.

3. PROPOSED SYSTEM

The proposed system is an intelligent air pollution monitoring and prediction platform that integrates machine learning techniques with real-time data acquisition and visualization. The system is designed to provide users with accurate Air Quality Index (AQI) information, pollutant analysis, and health recommendations through an interactive web-based application.

The system operates by collecting real-time air quality data from external APIs such as OpenWeatherMap. These APIs provide pollutant parameters including PM2.5, PM10, nitrogen dioxide (NO₂), carbon monoxide (CO), sulfur dioxide (SO₂), and ozone (O₃). The collected data is used both for visualization and as input to the machine learning model for prediction purposes.

Before training the model, the dataset undergoes preprocessing steps to ensure data quality and consistency. This includes removing irrelevant columns, handling missing values, and selecting only numerical features suitable for machine learning algorithms. These preprocessing steps are essential for improving the performance and reliability of the predictive model.

The core of the system is based on a Random Forest Regressor, which is trained using historical air quality data to predict AQI values. This model is chosen due to its ability to handle nonlinear relationships and provide accurate predictions. The model is evaluated using performance metrics such as Mean Absolute Error (MAE) and R² score to ensure its effectiveness.

The trained model is integrated into a Streamlit-based web application, allowing users to interact with the system in real time. Users can either enter a city name or select a location through an interactive map to retrieve air quality information. The system then displays both the live AQI values obtained from the API and the predicted AQI values generated by the machine learning model.

To enhance user understanding, the system provides graphical visualization of pollutant levels using bar charts, where the most dominant pollutant is highlighted.

Additionally, a map-based interface is used to display the selected geographical location, improving spatial awareness of pollution levels.

The system also includes a health advisory module that categorizes AQI values into levels such as Good, Moderate, Poor, and Severe. Based on these categories, appropriate health recommendations are provided to help users take preventive measures and reduce exposure to harmful pollutants.

Overall, the proposed system offers a comprehensive solution for real-time air quality monitoring and prediction. By combining machine learning, API integration, and interactive visualization, it provides an effective tool for increasing public awareness and supporting data-driven decision-making related to environmental health.

3.1 ARCHITECTURE DIAGRAM:

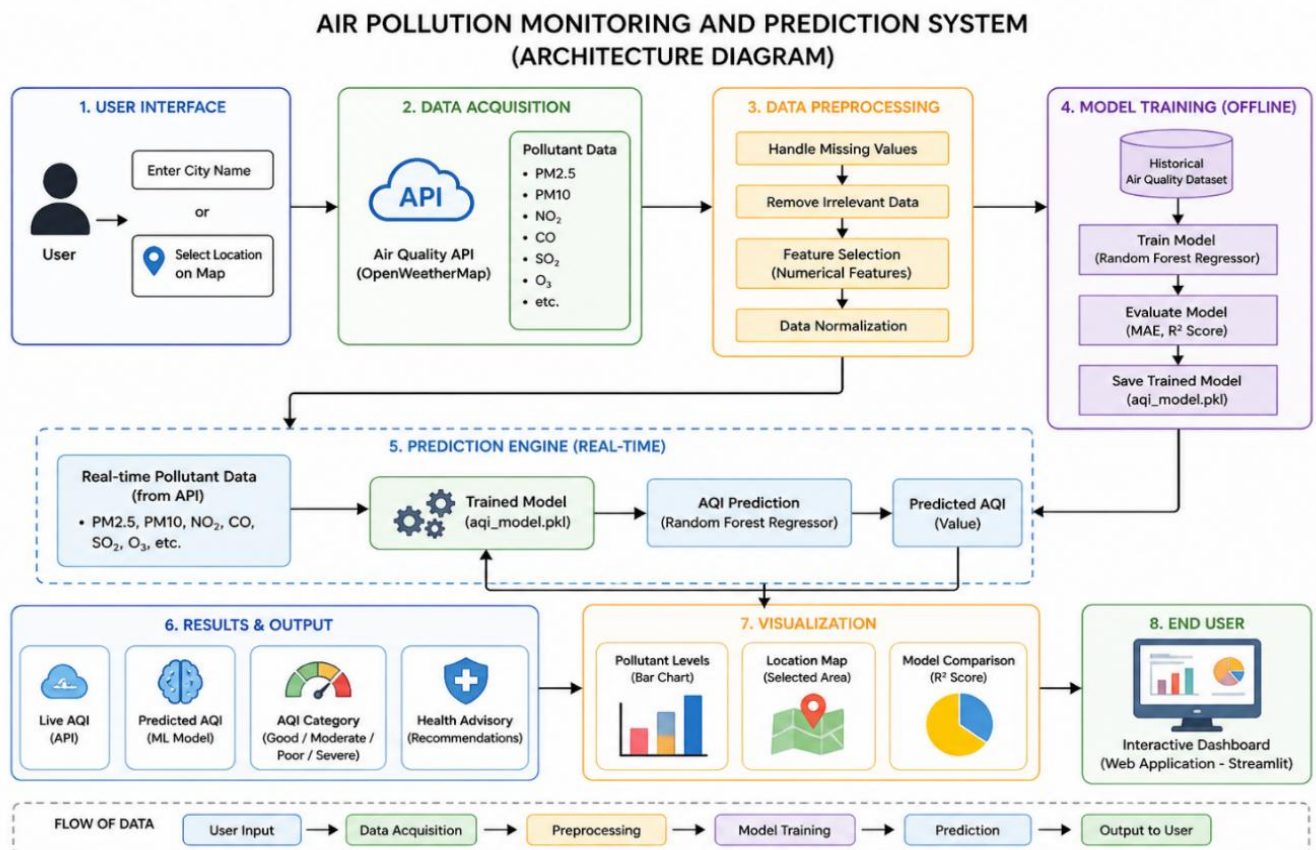


Figure 1: System Architecture Diagram

4. MODULES DESCRIPTION

4.1 MODULE 1 – Air Pollution Data Acquisition and Processing

The first module of the proposed system focuses on collecting and processing air pollution data required for analysis and prediction. This module is responsible for retrieving real-time pollutant information from external sources such as the OpenWeatherMap API. When a user enters a city name or selects a location on the map, the system sends a request to the API and obtains pollutant parameters including PM2.5, PM10, nitrogen dioxide (NO₂), carbon monoxide (CO), sulfur dioxide (SO₂), and ozone (O₃).

Once the data is collected, it undergoes preprocessing steps to ensure quality and consistency. These steps include handling missing values, converting data into numerical format, and organizing it into a structured form suitable for machine learning input. The preprocessing stage removes inconsistencies and prepares the dataset for further analysis. This module plays a crucial role in ensuring that accurate and reliable data is provided to the prediction model.

4.2 MODULE 2 – Machine Learning Model Training and Prediction

The second module focuses on building and utilizing a machine learning model for AQI prediction. In this module, historical air quality data is used to train a Random Forest Regressor model. The model learns the relationship between pollutant concentrations and AQI values by analyzing patterns in the dataset. During

training, the dataset is divided into training and testing sets to evaluate model performance using metrics such as Mean Absolute Error (MAE) and R^2 score. Once trained, the model is saved and integrated into the application. During real-time operation, the preprocessed pollutant data obtained from the API is passed into the trained model, which predicts the AQI value. This module forms the core intelligence of the system, enabling it to provide accurate and data-driven AQI predictions.

4.3 MODULE 3 – Visualization and User Interface

The third module focuses on presenting the results to the user through an interactive and user-friendly interface. The system is implemented using Streamlit, which allows users to interact with the application easily. Users can either enter a city name or select a location using an interactive map to obtain air quality information.

The module displays both live AQI values obtained from the API and predicted AQI values generated by the machine learning model. Additionally, it provides graphical visualization of pollutant levels using bar charts, where the most dominant pollutant is highlighted for better understanding. A map visualization is also included to show the selected location, enhancing spatial awareness. This module ensures that complex data is presented in a simple and understandable manner.

4.4 MODULE 4 – Health Advisory System

The final module focuses on providing health recommendations based on AQI levels. The predicted AQI values are categorized into different levels such as Good, Moderate, Poor, Very Poor, and Severe. Based on these categories, the system generates appropriate health advisories to guide users in taking preventive measures.

For example, when AQI levels are low, users are informed that outdoor activities are safe. When AQI levels are high, the system recommends limiting outdoor exposure, wearing masks, or staying indoors. This module enhances the practical usefulness of the system by not only providing data but also offering actionable insights for user safety.

5. IMPLEMENTATION AND RESULTS

5.1 IMPLEMENTATION

The implementation of the proposed Air Pollution Monitoring and Prediction System was carried out using Python and modern data science and web development frameworks. The system integrates machine learning, real-time data acquisition, and interactive visualization to provide an efficient solution for air quality analysis. Python was used as the primary programming language due to its extensive

ecosystem of libraries for data processing, machine learning, and web application development.

The machine learning model was developed using the Scikit-learn library, where a Random Forest Regressor was trained on historical air quality datasets. The dataset included pollutant parameters such as PM_{2.5}, PM₁₀, NO₂, CO, SO₂, and O₃, which were used as input features for predicting the Air Quality Index (AQI). Before training, the dataset underwent preprocessing steps including removal of irrelevant attributes, handling of missing values, and selection of numerical features to ensure data consistency and accuracy.

The dataset was divided into training and testing sets to evaluate model performance. The model was trained on the training data and tested using unseen data to assess its predictive capability. Performance evaluation was conducted using metrics such as Mean Absolute Error (MAE) and R² score. After successful training, the model was saved using joblib and integrated into the application for real-time prediction.

The system was deployed as a web application using Streamlit, which provides an interactive and user-friendly interface. Users can enter a city name or select a location using a map interface to retrieve air quality data. The application fetches real-time pollutant data from the OpenWeatherMap API and uses it as input for both

visualization and machine learning prediction. The integration of API data with the trained model enables real-time AQI prediction.

The application also includes visualization components such as bar charts to represent pollutant levels and a map to display the selected location. Additionally, a health advisory system categorizes AQI values and provides recommendations to users based on pollution levels. The overall implementation ensures smooth interaction between data collection, prediction, and user interface components.

5.2 RESULTS

The implemented system successfully provides real-time air quality monitoring and accurate AQI prediction based on pollutant data. Experimental results show that the Random Forest model performs effectively in capturing relationships between pollutant concentrations and AQI values, resulting in reliable predictions.

When users input different city names or select locations on the map, the system retrieves corresponding pollutant data and displays both live AQI values and predicted AQI values. The predictions are generally consistent with real-world pollution conditions, demonstrating the effectiveness of the trained model.

The visualization module effectively represents pollutant levels through bar charts, allowing users to identify dominant pollutants in a given area. The map-based interface enhances user interaction by providing a spatial understanding of air

quality conditions. The health advisory system further improves usability by offering clear and actionable recommendations based on AQI categories.

Testing of the system confirmed that it operates efficiently in real time, providing quick responses to user inputs. The integration of machine learning with real-time data and visualization makes the system practical and user-friendly. Overall, the results demonstrate that the proposed system successfully combines data analysis, machine learning, and web technologies to provide an effective air quality monitoring and prediction solution.

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